THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET B

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The transfer of heat from the hotter part of an object to the colder part is: 1

(a) convection **(b)** conduction **(c)** radiation **(d)** evaporation

2. Which of these is a good conductor of heat? 1

(a) Plastic **(b)** Wood **(c)** Air **(d)** Aluminium

3. Two thin blankets feel warmer than one thick one because of: 1

(a) weight **(b)** air trapped between them

(c) colour **(d)** smell

4. A sea breeze blows: 1

(a) from sea to land during the day **(b)** from land to land

(c) only at night **(d)** from land to sea during the day

5. A radiator or room heater warms nearby objects mostly by: 1

(a) convection only **(b)** radiation **(c)** evaporation **(d)** conduction

6. Water changes into vapour by: 1

(a) freezing **(b)** infiltration **(c)** evaporation **(d)** condensation

7. Underground layers of sediments and rocks that store water are: 1

(a) glaciers **(b)** dunes **(c)** estuaries **(d)** aquifers

8. A clinical thermometer measures temperature between: 1

(a) 100 °C and 200 °C **(b)** 35 °C and 42 °C

(c) 0 °C and 100 °C **(d)** −10 °C and 50 °C

9. A thermos flask keeps liquids hot by reducing: 1

(a) only light **(b)** only conduction

(c) conduction, convection and radiation **(d)** none

10. The Sun, ocean and wind drive: 1

(a) the water cycle **(b)** volcanoes **(c)** tides only **(d)** earthquakes

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Why are handles of cooking vessels made of wood or plastic? 2

12. What is a sea breeze? 2

13. Why do we prefer light-coloured clothes in summer? 2

14. Define evaporation and condensation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How does a thermos flask keep contents hot or cold? 3

16. Why do we feel hotter in dark clothes in summer? 3

17. What are the effects of groundwater depletion and how can we recharge it? 3

OR

How does aquifer storage help in times of drought? 3

18. Why are hot-air balloons able to rise? 3

OR

State three ways to keep a house cool in summer. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Why are some materials good conductors? 5
(ii) List examples.
(iii) State uses of insulators.

OR

(i) Describe an experiment to show convection. 5
(ii) Draw the set-up (describe).
(iii) State the observation.

20. **Case study:** A village has a dry hand pump after summer. 5
(i) What is the reason?
(ii) How can aquifers be recharged?
(iii) What is infiltration?
(iv) Name a practice.

OR

(i) How do clothes colour affect comfort? 5
(ii) Explain with absorption and reflection.
(iii) Give practical examples.

END OF QUESTION PAPER — Total: 40 Marks

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